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Формула:

Экз субъективный

$$\frac{0,3 \times \text{ЭКЗ} + (0,3 \times \text{БАЗ} + 0,25 \times \text{МАЗ} + 0,15 \times \text{ПР})}{0,7} \leftarrow \text{округл. сверху} \geq 8$$

Plan:

- Backpropagation
 - fully connected neural nets
-

$$a(x) = w_1 x_1 + \dots + w_n x_n = f(w, x)$$

↑
input

$$x \in \mathbb{R}^n$$

$w \in W$ — weights

$\{(x_i, y_i)\}_{i=1}^l$ training set

$Q(\hat{y}, y)$ - loss function

$$\mathcal{L}(\omega, \dots) = \frac{1}{l} \sum_{i=1}^l Q(f(\omega, x_i), y_i)$$

$\rightarrow \min_{\omega}$

Backpropagation

1. Chain rule

$$\frac{d(f(g(x)))}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx}$$

2. $y = f(g_1(x), g_2(x), \dots, g_n(x))$

$$\frac{dy}{dx} = \sum_{i=1}^n \frac{df}{dg_i} \cdot \frac{dg_i}{dx}$$

Backprop alternatives

- Historic:
- Evolutionary algorithms 1950s
 - Reservoir computing (2000)

Modern:

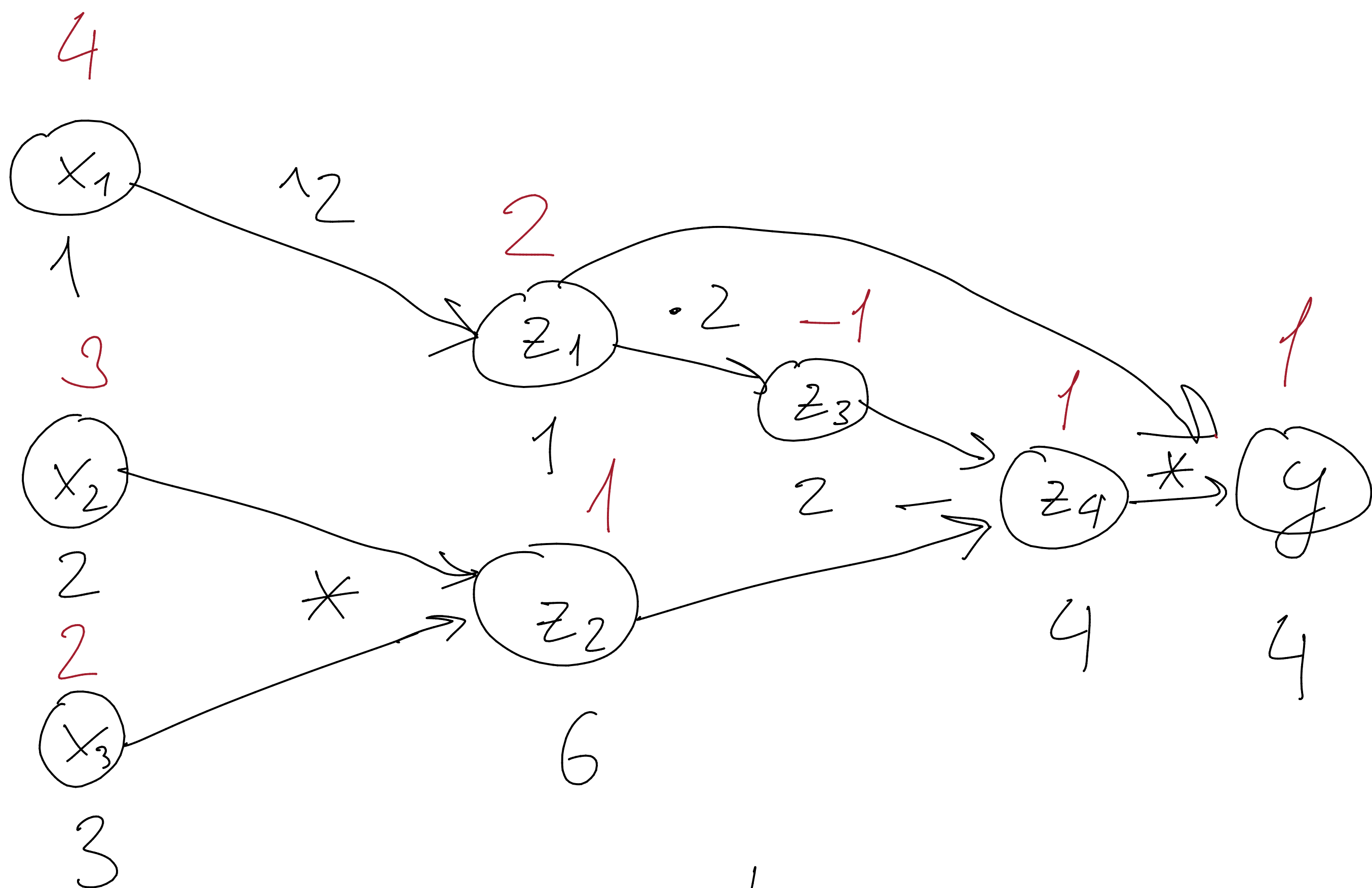
- Forward-forward (2022) pos
neg
- Direct feedback alignment (2016)

$$\left. \frac{d(f(g(x)))}{dx} \right|_{x_0} = \left. \frac{df}{dg} \right|_{g(x_0)} \cdot \left. \frac{dg}{dx} \right|_{x_0}$$

$\uparrow$
 x' - random
fixed

Computational graph

$$x_1^2 \cdot (x_2 x_3 - 2x_1^2) = f(x_1, x_2, x_3) = y$$



$$z_1 = x_1^2$$

$$z_2 = x_2 x_3$$

$$z_3 = 2z_1$$

$$z_4 = z_2 - z_3$$

$$y = z_1 z_4$$

$$\frac{dy}{dx} = ?$$

$$x_0 = (1, 2, 3)^T$$

$$\frac{dy}{dz_1} = (z_1 z_4)'_{z_1} = \frac{dz_1}{dz_1} \cdot z_4 + z_4'_{z_1}$$

$$\frac{dy}{dy} = 1$$

$$\frac{dy}{dz_4} = \frac{dy}{dy} \cdot \frac{dy}{dz_4} =$$

$$= 1 \cdot z_1 = 1$$

||
1

$$x^2 \rightarrow 2x \quad \forall x$$

$$x^2 \quad f(x) := 2x \rightarrow (x^2)' \Big|_2 = 4$$

$$\frac{dy}{dz_3} = \frac{dy}{dz_4} \cdot \frac{dz_4}{dz_3} = z_1 \cdot (-1) =$$

$$= 1 \cdot (-1) = -1$$

$$\frac{dy}{dz_2} = \frac{dy}{dz_4} \cdot \frac{dz_4}{dz_2} = 1 \cdot 1 = 1$$

$$\frac{dy}{dz_1} = \frac{dy}{dz_3} \cdot \frac{dz_3}{dz_1} + z_4 \cdot 1 =$$

$$= -1 \cdot 2 + 4 \cdot 1 = 2$$

$$\frac{dy}{dx_1} = \frac{dy}{dz_1} \cdot \frac{dz_1}{dx_1} = 2$$

$$\bullet 2x_1 = 4$$

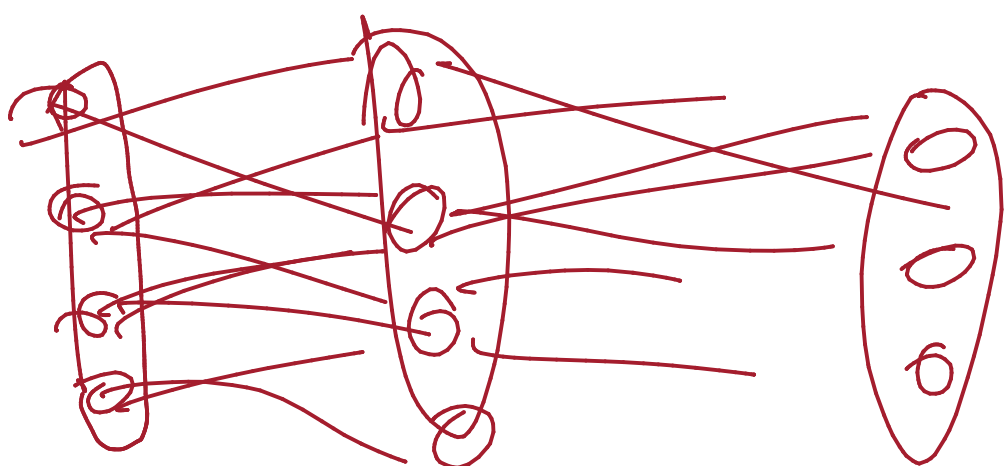
$$\frac{dg}{dx_2} = \frac{dg}{dz_2} \cdot \frac{dz_2}{dx_2} = 1 \cdot x_3 =$$

$$= 1 \cdot 3 = 3$$

$$\frac{dg}{dx_3} = \frac{dg}{dz_2} \cdot \frac{dz_2}{dx_3} = 1 \cdot x_2 =$$

$$= 1 \cdot 2 = 2$$

input



output

$$z_1 = f_1(x, w_1)$$

$$z_2 = f_2(z_1, w_2)$$

...

$$z_n = f_n(z_{n-1}, w_n)$$

$$\hat{y} = g(z_n)$$

$$l = Q(\hat{y}, y)$$

$$x \in \mathbb{R}^d \quad x = (x_1, \dots, x_n, 1)$$

$$a(x) = \langle w, x \rangle = w^T x$$

$$w \in \mathbb{R}^d$$

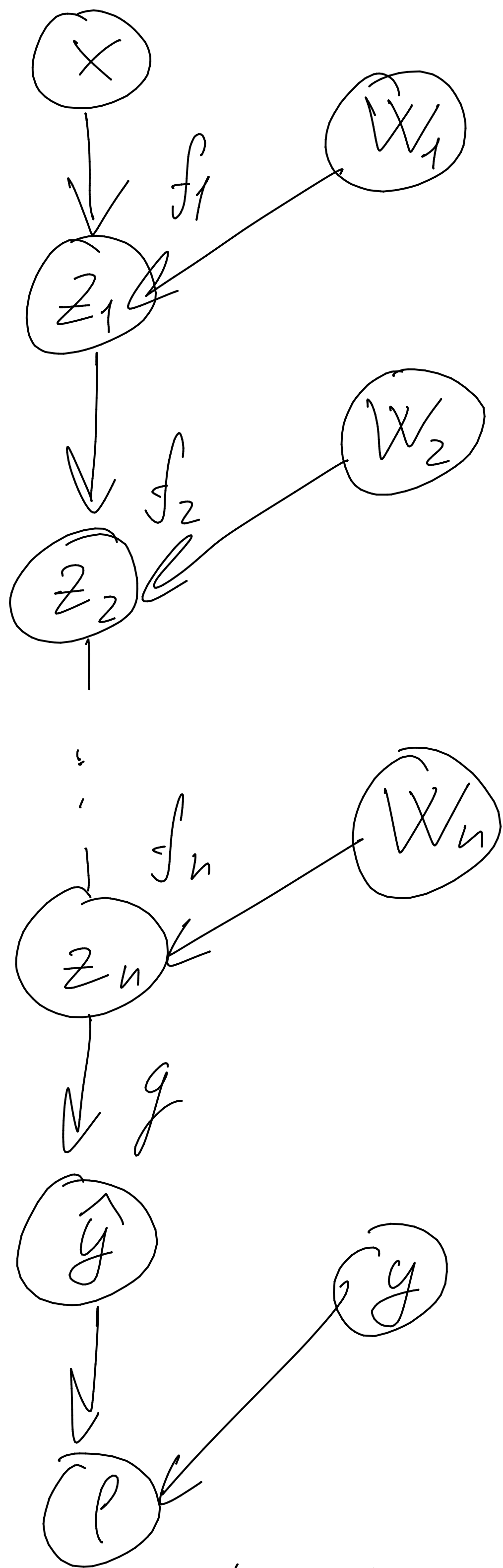
$$z_1 = (a_1(x), \dots, a_n(x))$$

$$= W_1 x + b_1$$

$$W_1 \in \mathbb{R}^{d_1 \times d}$$

$$z_2 = W_2 z_1 + b_2$$

$$d_1 \begin{pmatrix} \overbrace{w}^d \\ \overbrace{w}^d \\ \overbrace{w}^d \end{pmatrix} = W$$



,

$$W_2 \in \mathbb{R}^{d_2 \times d_1}$$

$$z_n = W_n z_{n-1} + b_n \quad W_n \in \mathbb{R}^{d_n \times d_{n-1}}$$

$$z_n = \underbrace{W_n \dots W_2 W_1}_{W \in \mathbb{R}^{d_n \times d_0}} x + b$$

$$\sigma(\cdot) : \mathbb{R} \rightarrow \mathbb{R} \quad \text{activation}$$

non-linear

func

differentiable

$$z_i = \sigma \left(\underbrace{W_i z_{i-1} + b_i}_{\text{dense / linear / fc layer}} \right)$$

dense / linear / fc layer

activation

z_i — latent representation
layer activations

$$z_i = (z_i^{(1)}, \dots, z_i^{(d_i)})$$

↑
neuron

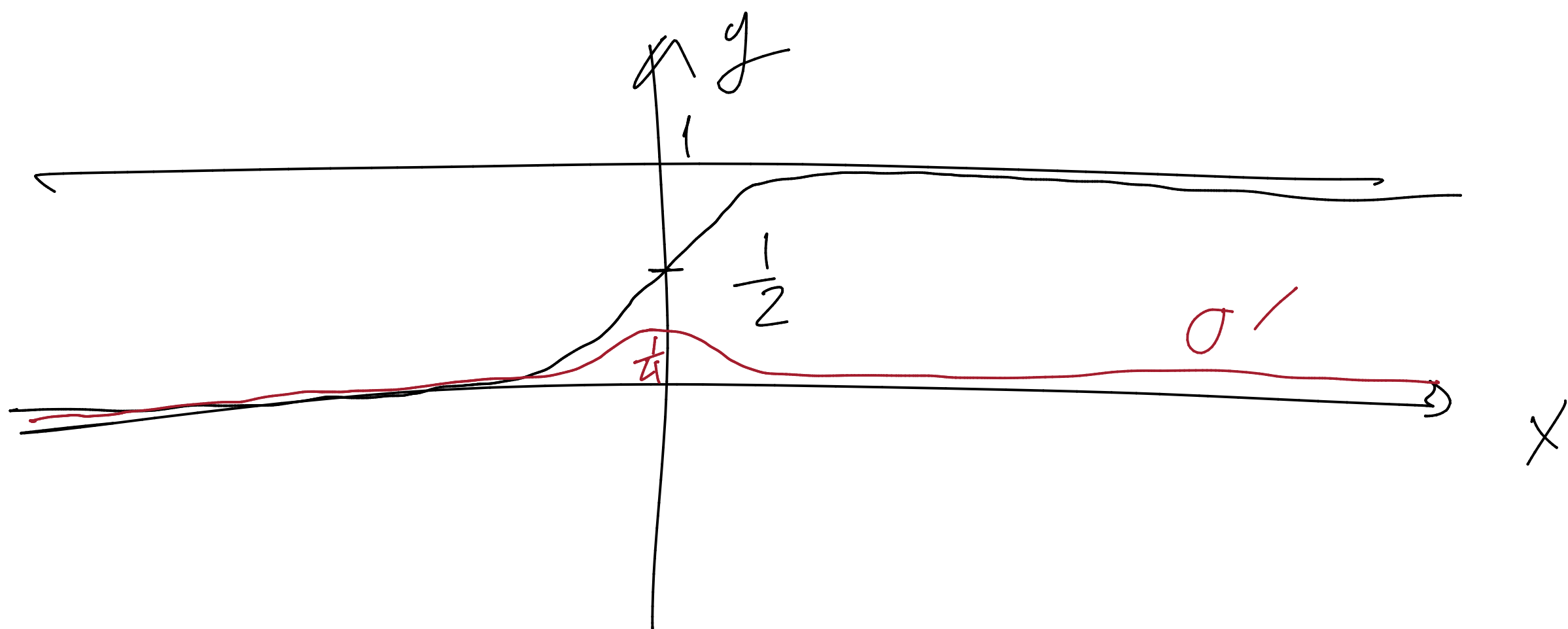
MLP Multi-layer Perceptron

FCN Fully Connected Network

1. Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$u_i = W_i z_i + b_i$$

$$z_{i+1} = \sigma(u_i)$$

$$\frac{dl}{dz_i} = \frac{dl}{dz_{i+1}}$$

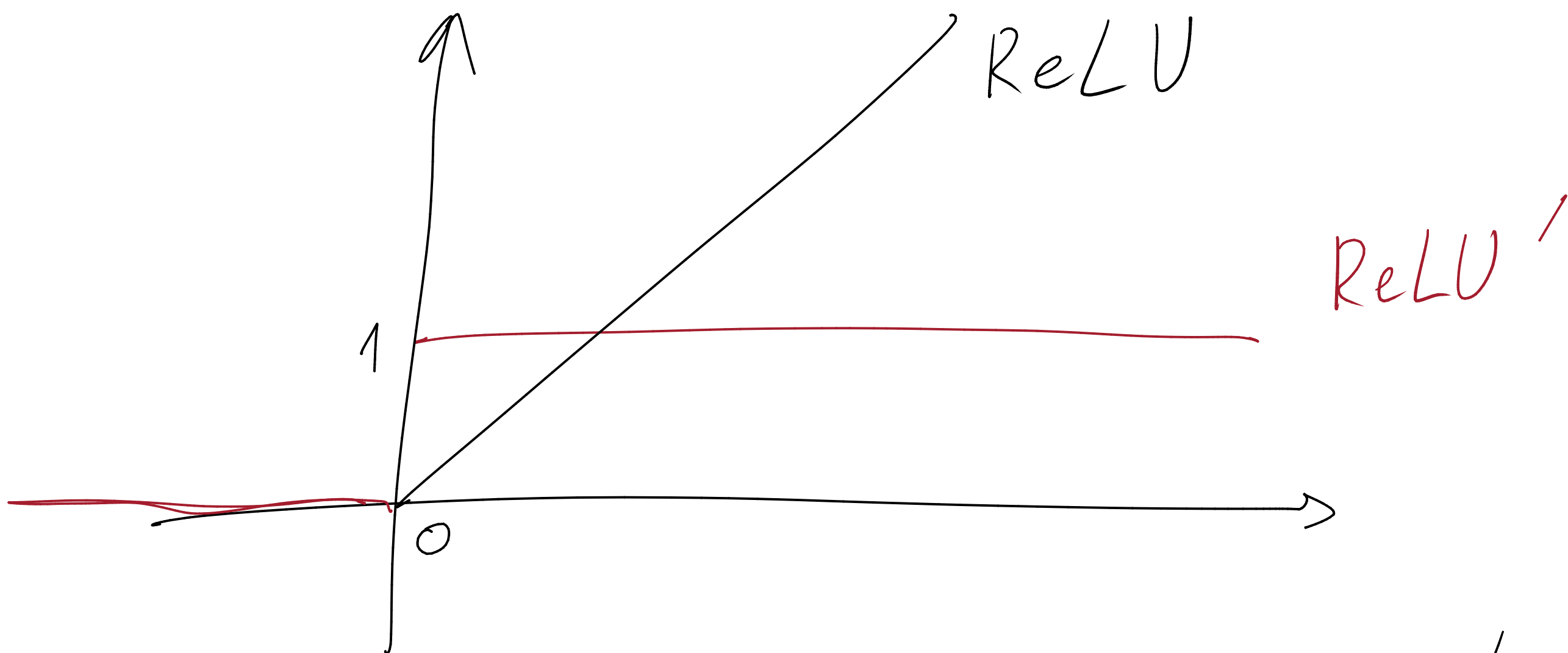
$$\bullet \sigma'(u_i) \cdot \frac{du_i}{dz_i}$$

$$|u_i| \uparrow \rightarrow \sigma'(u_i) \rightarrow 0$$

Vanishing gradients

2. ReLU (rectified linear unit)

$$\text{ReLU} = \max(0, x)$$



$$x \leq 0 \quad \text{grad} = 0$$

3. Leaky ReLU $(x) = \max(\alpha x, x)$

$$\alpha = 0, 1$$

